

44.5 Structured additive regression

Four kinds of term have appeared: a smooth function of a covariate, a coefficient varying with a modifier, an effect attached to a region of a map, an effect attached to a cluster. Each was a pair $(\mathbf{Z}_j, \mathbf{P}_j)$ — a design block and a nonnegative definite penalty — estimated by the same penalized criterion. This section says that out loud, shows that the framework contains the mixed models of [Chapter 32](#) exactly, and takes from them the thing still missing: a principled way to choose the smoothing parameters.

44.5.1 One predictor

Definition 44.5.1 — Structured additive regression

A structured additive predictor is

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \sum_{j=1}^J \mathbf{Z}_j \boldsymbol{\gamma}_j, \quad (44.5.1)$$

where $\mathbf{X}$ collects the unpenalized columns and each term j is a pair $(\mathbf{Z}_j, \mathbf{P}_j)$ with $\mathbf{Z}_j$ an $n \times d_j$ design block and $\mathbf{P}_j$ a symmetric nonnegative definite $d_j \times d_j$ penalty matrix, identified by the constraint $\mathbf{1}^\top \mathbf{Z}_j \boldsymbol{\gamma}_j = 0$ of Proposition 44.1.1(c). A **structured additive regression model** (STAR) fits [Eq. 44.5.1](#) by minimizing the penalized criterion [Eq. 44.1.4](#), with one smoothing parameter $\lambda_j \geq 0$ per term.

The content of the definition is the catalogue of pairs. Everything this chapter and [Chapter 43](#) have built is in [Table 44.5.1](#), and the list is open: a new kind of effect enters as soon as somebody writes down its design and its penalty.

Table 44.5.1. The catalogue of terms. Each row is a design block and a penalty; the last column is the dimension of the part of the term that the penalty never touches, which is what survives as $\lambda_j \rightarrow \infty$ and which the centring constraint reduces by one. In the last row m_1 and m_2 are the null-space dimensions of the two marginal penalties, so 1 for first differences in both coordinates and 4 for second differences.

term	design block $\mathbf{Z}_j$	penalty $\mathbf{P}_j$	$\dim \mathcal{N}(\mathbf{P}_j)$
polynomial of degree d	powers of z	$\mathbf{0}$	$d + 1$
P-spline, difference order r	B-splines at z	$\mathbf{D}_r^\top \mathbf{D}_r$	r
smoothing spline	natural spline basis	$\int B_l'' B_m''$	2
varying coefficient	B-splines times u	as for the P-spline	as for the P-spline
Markov random field	region indicators	Eq. 44.4.2	components of the map
kriging term	radial basis at knots	$\{r(\ \kappa_l - \kappa_m\)\}$	0
random intercept	cluster indicators	$\mathbf{I}$	0
random slope	cluster indicators times u	$\mathbf{I}$	0

term	design block $\mathbf{Z}_j$	penalty $\mathbf{P}_j$	$\dim \mathcal{N}(\mathbf{P}_j)$
tensor-product surface	products of B-splines	Eq. 44.3.3	$m_1 m_2$

Proposition 44.5.1 — When a STAR model is identified

In Definition 44.5.1, assume every $\mathbf{Z}_j$ has full column rank. The penalized criterion has a unique minimizer for every $\lambda_j > 0$ if and only if the only vector $\boldsymbol{\gamma} = (\boldsymbol{\beta}^\top, \boldsymbol{\gamma}_1^\top, \dots)^\top$ with $\mathbf{X}\boldsymbol{\beta} + \sum_j \mathbf{Z}_j \boldsymbol{\gamma}_j = \mathbf{0}$ and $\mathbf{P}_j \boldsymbol{\gamma}_j = \mathbf{0}$ for every j is $\boldsymbol{\gamma} = \mathbf{0}$. Equivalently: there is no exact concurrency among $\mathcal{C}(\mathbf{X})$ and the *unpenalized parts* $\mathbf{Z}_j \mathcal{N}(\mathbf{P}_j)$ of the terms.

Proof. The criterion is the convex quadratic $\|\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}\|^2 + \boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma}$ with $\mathbf{P}$ the block penalty Eq. 44.1.5, and its stationarity condition is Eq. 44.1.6 with $\mathbf{A} = \mathbf{Z}^\top \mathbf{Z} + \mathbf{P}$. If $\mathbf{A}$ is positive definite the criterion is strictly convex and Theorem 30.1.1 gives a unique minimizer. Conversely, if $\mathbf{A}$ is singular, pick $\boldsymbol{\gamma}_0 \neq \mathbf{0}$ with $\mathbf{A}\boldsymbol{\gamma}_0 = \mathbf{0}$; expanding the criterion at a minimizer $\hat{\boldsymbol{\gamma}}$ gives $Q(\hat{\boldsymbol{\gamma}} + t\boldsymbol{\gamma}_0) = Q(\hat{\boldsymbol{\gamma}}) + t^2 \boldsymbol{\gamma}_0^\top \mathbf{A}\boldsymbol{\gamma}_0 = Q(\hat{\boldsymbol{\gamma}})$ for every t , so the minimizer is not unique. Uniqueness for every $\lambda_j > 0$ is therefore positive definiteness of $\mathbf{A}$, that is $\boldsymbol{\gamma}^\top \mathbf{A}\boldsymbol{\gamma} = \|\mathbf{Z}\boldsymbol{\gamma}\|^2 + \sum_j \lambda_j \boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j = 0$ forces $\boldsymbol{\gamma} = \mathbf{0}$. Both summands are nonnegative, so the condition is exactly the one stated, and $\boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j = 0$ is equivalent to $\mathbf{P}_j \boldsymbol{\gamma}_j = \mathbf{0}$ by Theorem 1.7.4. ■

Two P-spline terms can therefore be identified even when their spaces nearly intersect, because only their *linear* parts — the null spaces of the second-difference penalties — need be independent. That is far weaker than the requirement for the unpenalized model in Proposition 44.2.3(a), and it explains why penalized additive fits rarely fail outright: they fail softly, through inflated variances.

44.5.2 The mixed-model representation

Proposition 44.5.2 — A STAR model is a linear mixed model

Let $\mathbf{P}_j$ have rank r_j and spectral decomposition $\mathbf{P}_j = \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j \mathbf{Q}_j^{+\top}$ with $\boldsymbol{\Lambda}_j = \text{diag}(\nu_{j1}, \dots, \nu_{jr_j})$ positive, and let $\mathbf{Q}_j^0$ be an orthonormal basis of $\mathcal{N}(\mathbf{P}_j)$. Put

$$\mathbf{X}_j^0 = \mathbf{Z}_j \mathbf{Q}_j^0, \quad \mathbf{Z}_j^+ = \mathbf{Z}_j \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j^{-1/2}, \quad \mathbf{u}_j = \boldsymbol{\Lambda}_j^{1/2} \mathbf{Q}_j^{+\top} \boldsymbol{\gamma}_j. \quad (44.5.2)$$

- (a) $\mathbf{Z}_j \boldsymbol{\gamma}_j = \mathbf{X}_j^0 \mathbf{b}_j + \mathbf{Z}_j^+ \mathbf{u}_j$ with $\mathbf{b}_j = \mathbf{Q}_j^{0\top} \boldsymbol{\gamma}_j$, and $\boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j = \|\mathbf{u}_j\|^2$.
- (b) With $\tau_j^2 = \sigma^2 / \lambda_j$, the penalized criterion Eq. 44.1.4 divided by σ^2 is the criterion Eq. 32.3.6 of Theorem 32.3.2 for the linear mixed model Eq. 32.2.1 with fixed design $[\mathbf{X}, \mathbf{X}_1^0, \dots, \mathbf{X}_J^0]$, random design $[\mathbf{Z}_1^+, \dots, \mathbf{Z}_J^+]$, $\mathbf{G} = \text{diag}(\tau_1^2 \mathbf{I}_{r_1}, \dots, \tau_J^2 \mathbf{I}_{r_J})$ and $\mathbf{R} = \sigma^2 \mathbf{I}$.
- (c) Consequently the penalized estimate is the BLUP of Theorem 32.3.1; the smoothing parameters are variance ratios, $\lambda_j = \sigma^2 / \tau_j^2$; and the restricted likelihood

Eq. 32.4.4 of the transformed model is a likelihood for $(\sigma^2, \tau_1^2, \dots, \tau_J^2)$, hence for $(\lambda_1, \dots, \lambda_J)$.

Proof.

- (a) The columns of $\mathbf{Q}_j^+$ and $\mathbf{Q}_j^0$ form an orthonormal basis of $\mathbb{R}^{d_j}$, so $\boldsymbol{\gamma}_j = \mathbf{Q}_j^0 \mathbf{b}_j + \mathbf{Q}_j^+ \mathbf{Q}_j^{+\top} \boldsymbol{\gamma}_j$ and $\mathbf{Q}_j^+ \mathbf{Q}_j^{+\top} \boldsymbol{\gamma}_j = \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j^{-1/2} \mathbf{u}_j$. Multiplying by $\mathbf{Z}_j$ gives the first claim. For the second, $\boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j = \boldsymbol{\gamma}_j^\top \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j \mathbf{Q}_j^{+\top} \boldsymbol{\gamma}_j = \|\mathbf{u}_j\|^2$.
- (b) Substituting (a) into Eq. 44.1.4 turns it into

$$\left\| \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \sum_j \mathbf{X}_j^0 \mathbf{b}_j - \sum_j \mathbf{Z}_j^+ \mathbf{u}_j \right\|^2 + \sum_j \lambda_j \|\mathbf{u}_j\|^2,$$

and dividing by σ^2 and writing $\lambda_j/\sigma^2 = 1/\tau_j^2$ gives $\mathbf{R}^{-1} = \sigma^{-2} \mathbf{I}$ in the first term and $\mathbf{G}^{-1}$ in the second, which is Eq. 32.3.6.

- (c) Immediate from Theorem 32.3.2(a)–(b), which identifies the minimizer of Eq. 32.3.6 with $(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}})$, and from Definition 32.4.1 applied to the transformed model. ■

Idea. A penalty is a prior, and a smoothing parameter is a variance ratio. That one identification does three jobs: the null space of a penalty becomes *fixed* effects, which is why Table 44.5.1 records $\dim \mathcal{N}(\mathbf{P}_j)$; it supplies a likelihood for $\boldsymbol{\lambda}$, which cross-validation does not; and it puts every result of Chapter 32 — BLUP, REML, Henderson’s equations, the posterior of Proposition 32.6.1 — at the service of smoothing.

44.5.3 Choosing the smoothing parameters

Two routes are now open. The first is prediction error: Definition 29.3.2 and the information criteria Proposition 29.2.4 apply verbatim with $\text{tr } \mathbf{S}$ in place of the number of parameters, minimized over $\boldsymbol{\lambda}$ numerically. The second is the restricted likelihood of Proposition 44.5.2(c), usually preferred: it is less prone to the occasional catastrophic undersmoothing cross-validation shows when its criterion is nearly flat, and it comes with the variance interpretation. Both choose from the data the fit uses, so Proposition 29.5.1 applies to both.

Maximizing the restricted likelihood directly is awkward, since each evaluation needs a determinant, and the updates below avoid it. What is proved is that they have the stated fixed points; that those are stationary points of the restricted likelihood is due to Schall (1991), with the general version in Wood and Fasiolo (2017).

Proposition 44.5.3 — Restricted-likelihood updates

In the mixed-model representation of Proposition 44.5.2, the expectation-maximization algorithm for the variance components, with $(\mathbf{b}, \mathbf{u})$ as the missing data, updates

$$\tau_j^{2,\text{new}} = \frac{1}{r_j} \left[\hat{\boldsymbol{\gamma}}_j^\top \mathbf{P}_j \hat{\boldsymbol{\gamma}}_j + \sigma^2 \text{tr}\{(\mathbf{A}^{-1})_{jj} \mathbf{P}_j\} \right], \quad (44.5.3)$$

where $\hat{\boldsymbol{\gamma}}_j$ and $\mathbf{A}$ are computed at the current $\boldsymbol{\lambda}$. A pair $(\boldsymbol{\tau}^2, \sigma^2)$ is a fixed point of Eq. 44.5.3 together with the update $\sigma^{2,\text{new}} = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 / (n - \text{df})$ if and only if

$$\hat{\tau}_j^2 = \frac{\hat{\boldsymbol{\gamma}}_j^\top \mathbf{P}_j \hat{\boldsymbol{\gamma}}_j}{\text{df}_j - m_j}, \quad \hat{\sigma}^2 = \frac{\|\mathbf{y} - \hat{\mathbf{y}}\|^2}{n - \text{df}}, \quad \hat{\lambda}_j = \frac{\hat{\sigma}^2}{\hat{\tau}_j^2}, \quad (44.5.4)$$

with $m_j = d_j - r_j = \dim \mathcal{N}(\mathbf{P}_j)$ and df_j , df the effective degrees of freedom of Proposition 44.1.2.

Proof. Eq. 44.5.3 is the expectation-maximization step for a variance component in a normal linear mixed model: the complete-data estimate of τ_j^2 is $\|\mathbf{u}_j\|^2 / r_j$, whose conditional expectation given $\mathbf{y}$ is $\|\hat{\mathbf{u}}_j\|^2 / r_j + \text{tr}\{\text{Cov}(\mathbf{u}_j \mid \mathbf{y})\} / r_j$. By Proposition 44.5.2(a), $\|\hat{\mathbf{u}}_j\|^2 = \hat{\boldsymbol{\gamma}}_j^\top \mathbf{P}_j \hat{\boldsymbol{\gamma}}_j$, and by Proposition 32.6.1(a) the conditional covariance of $\boldsymbol{\gamma}_j$ is $\sigma^2(\mathbf{A}^{-1})_{jj}$, so Eq. 44.5.2 gives

$$\text{tr Cov}(\mathbf{u}_j \mid \mathbf{y}) = \sigma^2 \text{tr}\{\boldsymbol{\Lambda}_j^{1/2} \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j^{-1} (\mathbf{A}^{-1})_{jj} \mathbf{Q}_j^+ \boldsymbol{\Lambda}_j^{1/2}\} = \sigma^2 \text{tr}\{(\mathbf{A}^{-1})_{jj} \mathbf{P}_j\},$$

the trace term as written. (The EM step itself is quoted from the mixed-model literature; what is proved here is its fixed point.)

For the fixed point, Proposition 44.1.2(b) gives $\text{df}_j = d_j - \lambda_j \text{tr}\{(\mathbf{A}^{-1})_{jj} \mathbf{P}_j\}$, that is

$$\text{tr}\{(\mathbf{A}^{-1})_{jj} \mathbf{P}_j\} = \frac{d_j - \text{df}_j}{\lambda_j} = \frac{(d_j - \text{df}_j) \tau_j^2}{\sigma^2}.$$

Substituting into Eq. 44.5.3 and demanding $\tau_j^{2,\text{new}} = \tau_j^2$,

$$r_j \tau_j^2 = \hat{\boldsymbol{\gamma}}_j^\top \mathbf{P}_j \hat{\boldsymbol{\gamma}}_j + (d_j - \text{df}_j) \tau_j^2,$$

so $\tau_j^2(r_j - d_j + \text{df}_j) = \hat{\boldsymbol{\gamma}}_j^\top \mathbf{P}_j \hat{\boldsymbol{\gamma}}_j$, and $r_j - d_j = -m_j$ gives the first equation of Eq. 44.5.4. The statement for σ^2 is its own update, and $\lambda_j = \sigma^2 / \tau_j^2$ is Proposition 44.5.2(c). ■

Read Eq. 44.5.4 as an estimate of a variance: the numerator is the roughness the term shows, the denominator $\text{df}_j - m_j$ the number of dimensions it used in showing it, discounting the unpenalized part the penalty cannot see. The solver is one assembly and one solve,

```
import numpy as np

def assemble(blocks, X, lambdas):
    """Stack the parametric columns and the terms into one design and one block
    penalty."""
    Z = np.column_stack([X] + [Zj for Zj, _ in blocks])
    K = np.zeros((Z.shape[1], Z.shape[1]))
    start = X.shape[1]
    for (_, Kj), lam in zip(blocks, lambdas):
        d = Kj.shape[0]
        K[start:start + d, start:start + d] = lam * Kj
        start += d
    return Z, K
```

```

class Fit:
    """One penalized solve: coefficients, components, operators and degrees of
    freedom."""

    def __init__(self, y, blocks, X, lambdas, W=None, z=None, operators=True):
        Z, K = assemble(blocks, X, lambdas)
        w = np.ones(len(y)) if W is None else W
        Zw = Z * w[:, None]
        B = Z.T @ Zw                                     # the weighted cross-
        product
        A = B + K
        Ainv = np.linalg.inv(A)
        self.Z, self.K, self.A, self.w = Z, K, A, w
        self.gamma = Ainv @ (Zw.T @ (y if z is None else z))
        self.fitted = Z @ self.gamma
        self.edf_total = np.trace(Ainv @ B)               # tr(S) without
        forming S
        self.parts, self.edf, self.comp = [], [], []
        start = X.shape[1]
        for Zj, _ in blocks:
            sl = slice(start, start + Zj.shape[1])
            self.parts.append(Zj @ self.gamma[sl])
            self.edf.append(np.trace(Ainv[sl] @ B[:, sl]))
            start += Zj.shape[1]
        if operators:                                     # the n x n smoothers
            C = Ainv @ Zw.T
            self.S = Z @ C
            start = X.shape[1]
            for Zj, _ in blocks:
                self.comp.append(Zj @ C[start:start + Zj.shape[1]])
                start += Zj.shape[1]
        self.beta = self.gamma[:X.shape[1]]

```

[Run this code online](#)

and the updates themselves are a handful more:

```

def null_dim(K, tol=1e-9):
    """The dimension of the penalty null space: the part of a term that is never
    penalized."""
    e = np.linalg.eigvalsh(K)
    return int(np.sum(e < tol * max(e.max(), 1.0)))

def reml_lambdas(y, blocks, X, lambdas=None, iters=80, tol=1e-8):
    """Restricted-likelihood updates for a normal response: refit, then set each
    variance component to  $\tau_j^{-2} = \gamma_j' K_j \gamma_j / (df_j - m_j)$  and  $\sigma^2$  to the
    penalized residual sum of squares over  $n - df$ , and put  $\lambda_j = \sigma^2 / \tau_j^{-2}$ ."""
    lam = np.ones(len(blocks)) if lambdas is None else np.asarray(lambdas, float)
    for _ in range(iters):
        fit = Fit(y, blocks, X, lam, operators=False)
        sigma2 = np.sum((y - fit.fitted) ** 2) / (len(y) - fit.edf_total)
        new = lam.copy()
        start = X.shape[1]
        for j, (_, Kj) in enumerate(blocks):
            d = Kj.shape[0]
            g = fit.gamma[start:start + d]

```

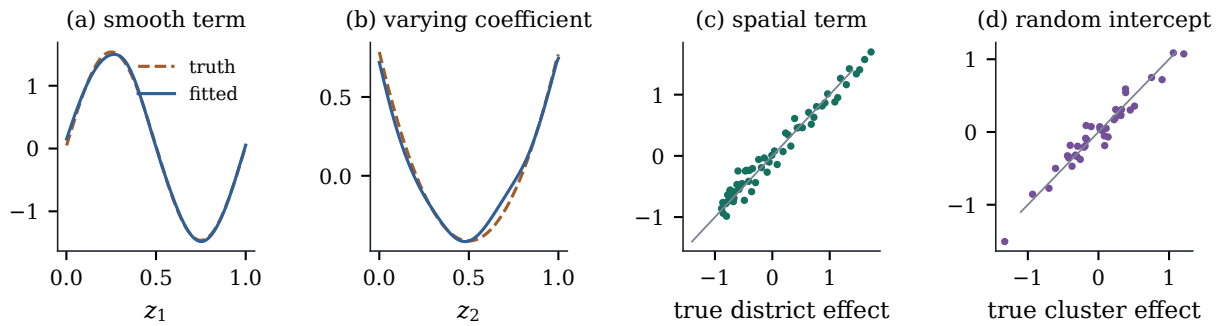


Figure 44.5.1: One structured additive fit with four kinds of term. (a) the smooth term against the truth. (b) the estimated coefficient function. (c) and (d) fitted district and cluster effects against the truth, with the line of equality; both are shrunk, as a best linear unbiased predictor must be.

```

tau2 = max(g @ Kj @ g, 1e-12) / max(fit.edf[j] - null_dim(Kj), 1e-6)
new[j] = min(max(sigma2 / tau2, 1e-8), 1e10)
start += d
done = np.max(np.abs(np.log(new) - np.log(lam))) < tol
lam = new
if done:
    break
return lam, Fit(y, blocks, X, lam), sigma2

```

[Run this code online](#)

Example. (Four kinds of term at once)

A simulated data set of 1200 observations has a binary covariate with a linear effect, a smooth effect of z_1 , a coefficient of u varying with z_2 , an areal effect on an 8×8 map of 64 districts and a random intercept for 40 clusters: four terms, 132 coefficients, smoothing parameters from [Eq. 44.5.4](#).

The fit recovers the residual standard deviation as 0.713 (true 0.7) and the linear effect as 0.770 (true 0.8). The four terms take 7.86, 5.13, 42.92 and 36.48 effective degrees of freedom, 94.40 out of 132, with implied variance components 0.1609, 0.0170, 0.2002 and 0.2646 — the last the variance of the random intercept, truly 0.25. Root mean squared errors against the truth are 0.031, 0.045, 0.129 and 0.119, against a noise standard deviation of 0.7. [Figure 44.5.1](#) shows all four.

The shrinkage in panels (c) and (d) is not a defect but [Theorem 32.3.1](#): a predictor of a random effect trades bias for variance, and the slope of predicted against true falls below one by exactly the amount the estimated variance ratio dictates.

44.5.4 The Bayesian reading

[Proposition 44.5.2](#) already contains it. Give each term the prior

$$\pi(\boldsymbol{\gamma}_j \mid \tau_j^2) \propto (\tau_j^2)^{-r_j/2} \exp\left\{-\frac{1}{2\tau_j^2} \boldsymbol{\gamma}_j^\top \mathbf{P}_j \boldsymbol{\gamma}_j\right\}, \quad (44.5.5)$$

normal on $\mathcal{N}(\mathbf{P}_j)^\perp$ and flat on the null space, give $\boldsymbol{\beta}$ a flat prior, and the penalized estimate is the posterior mode (Definition 39.4.1); with $\boldsymbol{\lambda}$ fixed the posterior of $\boldsymbol{\gamma}$ is normal with covariance $\sigma^2 \mathbf{A}^{-1}$ by Proposition 32.6.1(a). That covariance produces the bands in this chapter's figures: the Wahba–Nychka bands of Theorem 43.6.1, with their caveat that they are honest about bias averaged over the curve, not pointwise, and condition on $\boldsymbol{\lambda}$ as though it were known. Treating $\boldsymbol{\tau}^2$ and σ^2 as unknown too gives a fully Bayesian model, sampled as in Section 39.4 (Definition 39.4.2), whose advantage is precisely that it propagates the uncertainty in $\boldsymbol{\lambda}$ into the bands instead of ignoring it.

Warning (Improper priors on variance components). The prior Eq. 44.5.5 is improper on $\mathcal{N}(\mathbf{P}_j)$, which is harmless: that part is a fixed effect with a flat prior. The prior on τ_j^2 is not harmless. An inverse-gamma prior with both parameters tending to zero, or a flat prior on τ_j^2 , can produce an improper posterior, and the symptom in a sampler is not a crash but a chain that drifts towards zero variance and never settles (Proposition 39.5.1). Use a proper prior whose scale is set on the scale of the data, and check the sampler, never the fit alone.

44.5.5 Exercises

44.5.5.1 A. Check your understanding

Exercise A1.

For each of the following, give the design block and the penalty: (i) a random slope of a covariate u within K clusters; (ii) a linear effect that is not to be penalized at all; (iii) a smooth term that is to be penalized towards a straight line rather than towards a constant.

44.5.5.2 B. Practice

Exercise B1.

Take one term with $\mathbf{Z} = \mathbf{D}$ the $n \times K$ matrix of cluster indicators and $\mathbf{P} = \mathbf{I}$. Show that the penalized estimate of the cluster effects is $\hat{\gamma}_k = m_k \bar{r}_k / (m_k + \lambda)$, where m_k is the cluster size and $\bar{r}_k$ the mean residual in cluster k , and identify this with Theorem 32.3.1 for the one-way random effects model of Definition 32.1.1.

Solution. With the other effects fixed the criterion is $\sum_k \sum_{i \in k} (r_i - \gamma_k)^2 + \lambda \sum_k \gamma_k^2$, which separates over k and gives $\hat{\gamma}_k = m_k \bar{r}_k / (m_k + \lambda)$. With $\lambda = \sigma^2 / \tau^2$ that is $\bar{r}_k m_k \tau^2 / (m_k \tau^2 + \sigma^2)$, the shrinkage factor of Theorem 32.3.1 for the one-way model.

Exercise B2.

Write the generalized cross-validation criterion Definition 29.3.2 for a STAR fit as $\text{GCV}(\boldsymbol{\lambda}) = n \|\mathbf{y} - \mathbf{S}\mathbf{y}\|^2 / (n - \text{tr } \mathbf{S})^2$. Show that it is minimized where σ^2 of Eq. 44.5.4 is stationary in a particular sense, and explain why the two criteria nevertheless disagree in general.

44.5.5.3 C. Going deeper

Exercise C1.

The updates [Eq. 44.5.4](#) are a fixed-point iteration, not a descent method, and they are not guaranteed to converge from every start. Construct a two-term model in which the restricted likelihood has a stationary point at $\tau_1^2 = 0$ (the term is smoothed away), verify that this is a fixed point of [Eq. 44.5.4](#), and explain how a run that lands there should be reported.

Solution. Let the first term be a P-spline in a covariate with no effect and the second anything. If the data show no roughness in that term, $\hat{\gamma}_1^\top \mathbf{P}_1 \hat{\gamma}_1 \rightarrow 0$ while $\text{df}_1 \rightarrow m_1$, so [Eq. 44.5.4](#) sends $\tau_1^2 \rightarrow 0$ and $\lambda_1 \rightarrow \infty$, and there the update reproduces itself because the numerator stays zero. This is the boundary of the parameter space, where [Theorem 32.5.2](#) says the usual asymptotics for a test of $\tau_1^2 = 0$ fail; the honest report is “the term was smoothed to its null space”, not “significant at 0 degrees of freedom”.